

SEQUENCE:

A sequence is just a list of elements usually written in a row.

EXAMPLES:

1. 1, 2, 3, 4, 5, ...
2. 4, 8, 12, 16, 20, ...
3. 2, 4, 8, 16, 32, ...
4. 1, 1/2, 1/3, 1/4, 1/5, ...
5. 1, 4, 9, 16, 25, ...
6. 1, -1, 1, -1, 1, -1, ...

NOTE: The symbol “...” is called ellipsis, and reads “and so forth”

FORMAL DEFINITION:

A sequence is a function whose domain is the set of integers greater than or equal to a particular integer n_0 . Usually this set is the set of Natural numbers $\{1, 2, 3, \dots\}$ or the set of whole numbers $\{0, 1, 2, 3, \dots\}$.

NOTATION:

We use the notation a_n to denote the image of the integer n , and call it a term of the sequence. Thus

$$a_1, a_2, a_3, a_4, \dots, a_n, \dots$$

represent the terms of a sequence defined on the set of natural numbers N .

Note that a sequence is described by listing the terms of the sequence in order of increasing subscripts.

FINDING TERMS OF A SEQUENCE GIVEN BY AN EXPLICIT FORMULA:

An explicit formula or general formula for a sequence is a rule that shows how the values of a_k depends on k .

EXAMPLE:

Define a sequence $a_1, a_2, a_3, \dots$ by the explicit formula

$$a_k = \frac{k}{k+1} \quad \text{for all integers } k \geq 1$$

The first four terms of the sequence are:

$$a = \frac{1}{1+1} = \frac{1}{2}, a = \frac{2}{2+1} = \frac{2}{3}, a = \frac{3}{3+1} = \frac{3}{4}$$

$$\text{and fourth term is } a = \frac{4}{4+1} = \frac{4}{5}$$

EXAMPLE:

Write the first four terms of the sequence defined by the formula

$$b_j = 1 + 2^j, \text{ for all integers } j \geq 0$$

SOLUTION:

$$b_0 = 1 + 2^0 = 1 + 1 = 2$$

$$b_1 = 1 + 2^1 = 1 + 2 = 3$$

$$b_2 = 1 + 2^2 = 1 + 4 = 5$$

$$b_3 = 1 + 2^3 = 1 + 8 = 9$$

REMARK:

The formula $b_j = 1 + 2^j$, for all integers $j \geq 0$ defines an infinite sequence having infinite number of values.

EXERCISE:

Compute the first six terms of the sequence defined by the formula

$$C_n = 1 + (-1)^n \text{ for all integers } n \geq 0$$

SOLUTION:

$$C_0 = 1 + (-1)^0 = 1 + 1 = 2$$

$$C_1 = 1 + (-1)^1 = 1 + (-1) = 0$$

$$C_2 = 1 + (-1)^2 = 1 + 1 = 2$$

$$C_3 = 1 + (-1)^3 = 1 + (-1) = 0$$

$$C_4 = 1 + (-1)^4 = 1 + 1 = 2$$

$$C_5 = 1 + (-1)^5 = 1 + (-1) = 0$$

REMARK:

1) If n is even, then $C_n = 2$ and if n is odd, then $C_n = 0$

Hence, the sequence oscillates endlessly between 2 and 0.

2) An infinite sequence may have only a finite number of values.

EXAMPLE:

Write the first four terms of the sequence defined by

$$C_n = \frac{(-1)^n n}{n+1} \text{ for all integers } n \geq 1$$

SOLUTION:

$$C_1 = \frac{(-1)^1(1)}{1+1} = \frac{-1}{2}, C_2 = \frac{(-1)^2(2)}{2+1} = \frac{2}{3}, C_3 = \frac{(-1)^3(3)}{3+1} = \frac{-3}{4}$$

$$\text{And fourth term is } C_4 = \frac{(-1)^4(4)}{4+1} = \frac{4}{5}$$

REMARK: A sequence whose terms alternate in sign is called an alternating sequence.

EXERCISE:

Find explicit formulas for sequences with the initial terms given:

1) 0, 1, -2, 3, -4, 5, ...

SOLUTION:

$$a_n = (-1)^{n+1}n \text{ for all integers } n \geq 0$$

$$2) \quad 1 - \frac{1}{2}, \frac{1}{2} - \frac{1}{3}, \frac{1}{3} - \frac{1}{4}, \frac{1}{4} - \frac{1}{5}, \dots$$

SOLUTION:

$$b_k = \frac{1}{k} - \frac{1}{k+1} \quad \text{for all integers } n \geq 1$$

$$3) \quad 2, 6, 12, 20, 30, 42, 56, \dots$$

SOLUTION:

$$C_n = n(n+1) \quad \text{for all integers } n \geq 1$$

$$4) \quad 1/4, 2/9, 3/16, 4/25, 5/36, 6/49, \dots$$

SOLUTION:

$$\text{OR} \quad d_i = \frac{i}{(i+1)^2} \quad \text{for all integers } i \geq 1$$

$$d_j = \frac{j+1}{(j+2)^2} \quad \text{for all integers } j \geq 0$$

ARITHMETIC SEQUENCE:

A sequence in which every term after the first is obtained from the preceding term by adding a constant number is called an arithmetic sequence or arithmetic progression (A.P.)

The constant number, being the difference of any two consecutive terms is called the common difference of A.P., commonly denoted by “d”.

EXAMPLES:

1. $5, 9, 13, 17, \dots$ (common difference = 4)
2. $0, -5, -10, -15, \dots$ (common difference = -5)
3. $x + a, x + 3a, x + 5a, \dots$ (common difference = 2a)

GENERAL TERM OF AN ARITHMETIC SEQUENCE:

Let **a** be the first term and **d** be the common difference of an arithmetic sequence. Then the sequence is $a, a+d, a+2d, a+3d, \dots$

If a_i , for $i \geq 1$, represents the terms of the sequence then

$$a_1 = \text{first term} = a = a + (1-1)d$$

$$a_2 = \text{second term} = a + d = a + (2-1)d$$

$$a_3 = \text{third term} = a + 2d = a + (3-1)d$$

By symmetry

$$a_n = \text{nth term} = a + (n-1)d \quad \text{for all integers } n \geq 1.$$

EXAMPLE:

Find the 20th term of the arithmetic sequence

$$3, 9, 15, 21, \dots$$

SOLUTION:

Here a = first term = 3

d = common difference = $9 - 3 = 6$

n = term number = 20

a_{20} = value of 20th term = ?

Since $a_n = a + (n - 1) d$; $n \geq 1$

$$\begin{aligned}\therefore a_{20} &= 3 + (20 - 1) 6 \\ &= 3 + 114 \\ &= 117\end{aligned}$$

EXAMPLE:

Which term of the arithmetic sequence

4, 1, -2, ..., is -77

SOLUTION:

Here a = first term = 4

d = common difference = $1 - 4 = -3$

a_n = value of n th term = -77

n = term number = ?

Since

$$a_n = a + (n - 1) d \quad n \geq 1$$

$$\Rightarrow -77 = 4 + (n - 1) (-3)$$

$$\Rightarrow -77 - 4 = (n - 1) (-3)$$

$$\text{OR } \frac{-81}{-3} = n - 1$$

$$\begin{aligned}\text{OR } -3 \\ 27 = n - 1 \\ n = 28\end{aligned}$$

Hence -77 is the 28th term of the given sequence.

EXERCISE:

Find the 36th term of the arithmetic sequence whose 3rd term is 7 and 8th term is 17.

SOLUTION:

Let a be the first term and d be the common difference of the arithmetic sequence.

Then

$$a_n = a + (n - 1)d \quad n \geq 1$$

$$\Rightarrow a_3 = a + (3 - 1) d$$

$$\text{and } a_8 = a + (8 - 1) d$$

Given that $a_3 = 7$ and $a_8 = 17$. Therefore

$$7 = a + 2d \dots\dots\dots(1)$$

$$\text{and } 17 = a + 7d \dots\dots\dots(2)$$

Subtracting (1) from (2), we get,

$$10 = 5d$$

$$\Rightarrow d = 2$$

Substituting $d = 2$ in (1) we have

$$7 = a + 2(2)$$

which gives $a = 3$

Thus, $a_n = a + (n - 1) d$

$$a_n = 3 + (n - 1) 2 \quad (\text{using values of } a \text{ and } d)$$

Hence the value of 36th term is

$$\begin{aligned} a_{36} &= 3 + (36 - 1) 2 \\ &= 3 + 70 \\ &= 73 \end{aligned}$$

GEOMETRIC SEQUENCE:

A sequence in which every term after the first is obtained from the preceding term by multiplying it with a constant number is called a geometric sequence or geometric progression (G.P.)

The constant number, being the ratio of any two consecutive terms is called the common ratio of the G.P. commonly denoted by “r”.

EXAMPLE:

1. 1, 2, 4, 8, 16, ... (common ratio = 2)
2. 3, - 3/2, 3/4, - 3/8, ... (common ratio = - 1/2)
3. 0.1, 0.01, 0.001, 0.0001, ... (common ratio = 0.1 = 1/10)

GENERAL TERM OF A GEOMETRIC SEQUENCE:

Let **a** be the first term and **r** be the common ratio of a geometric sequence. Then the sequence is $a, ar, ar^2, ar^3, \dots$

If a_i , for $i \geq 1$ represent the terms of the sequence, then

$$\begin{aligned} a_1 &= \text{first term} = a = ar^{1-1} \\ a_2 &= \text{second term} = ar = ar^{2-1} \\ a_3 &= \text{third term} = ar^2 = ar^{3-1} \\ &\dots\dots\dots \\ &\dots\dots\dots \\ a_n &= \text{nth term} = ar^{n-1}; \text{ for all integers } n \geq 1 \end{aligned}$$

EXAMPLE:

Find the 8th term of the following geometric sequence

$$4, 12, 36, 108, \dots$$

SOLUTION:

$$\begin{aligned} \text{Here } a &= \text{first term} = 4 \\ r &= \text{common ratio} = \frac{12}{4} = 3 \\ n &= \text{term number} = 8 \\ a_8 &= \text{value of 8th term} = ? \end{aligned}$$

$$\text{Since } a_n = ar^{n-1}; \quad n \geq 1$$

$$\begin{aligned} \Rightarrow a_8 &= (4)(3)^{8-1} \\ &= 4(2187) \\ &= 8748 \end{aligned}$$

EXAMPLE:

Which term of the geometric sequence is 1/8 if the first term is 4 and common ratio $\frac{1}{2}$

SOLUTION:

$$\begin{aligned} \text{Given } a &= \text{first term} = 4 \\ r &= \text{common ratio} = 1/2 \end{aligned}$$

a_n = value of the nth term = $1/8$

n = term number = ?

Since $a_n = ar^{n-1}$ $n \geq 1$

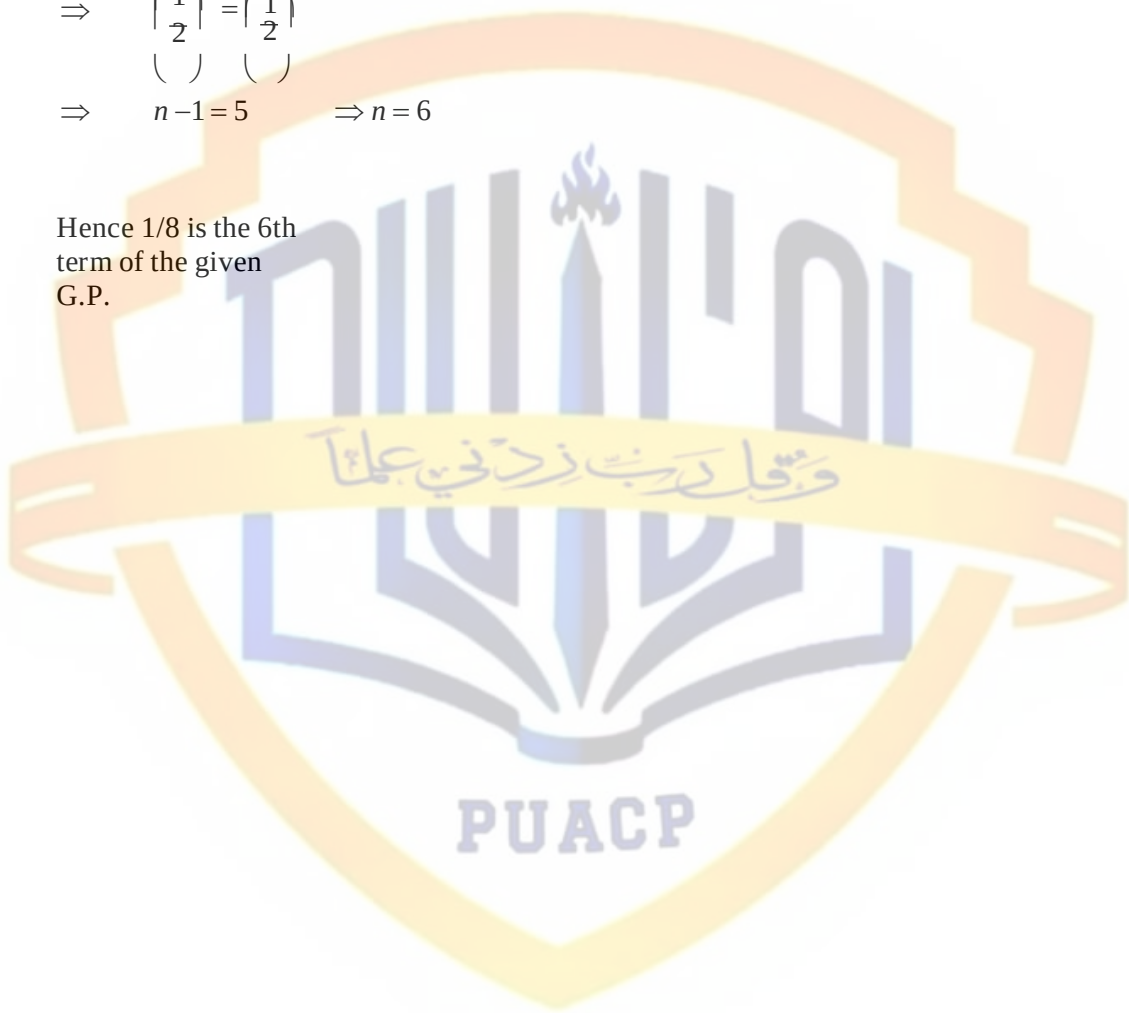
$$\Rightarrow \frac{1}{8} = 4 \left(\frac{1}{2} \right)^{n-1}$$

$$\Rightarrow \frac{1}{32} = \left(\frac{1}{2} \right)^{n-1}$$

$$\Rightarrow \left(\frac{1}{2} \right)^5 = \left(\frac{1}{2} \right)^{n-1}$$

$$\Rightarrow n-1 = 5 \Rightarrow n = 6$$

Hence $1/8$ is the 6th term of the given G.P.



SERIES:

The sum of the terms of a sequence forms a series. If $a_1, a_2, a_3, \dots$ represent a sequence of numbers, then the corresponding series is

$$a_1 + a_2 + a_3 + \dots = \sum_{k=1}^{\infty} a_k$$

SUMMATION NOTATION:

The capital Greek letter sigma Σ is used to write a sum in a short hand notation.

where k varies from 1 to n represents the sum given in expanded form by

$$= a_1 + a_2 + a_3 + \dots + a_n$$

More generally if m and n are integers and $m \leq n$, then the summation from k equal m to n of a_k is

$$\sum_{k=m}^n a_k = a_m + a_{m+1} + a_{m+2} + \dots + a_n$$

Here **k** is called the index of the summation; **m** the lower limit of the summation and **n** the upper limit of the summation.

COMPUTING SUMMATIONS:

Let $a_0 = 2$, $a_1 = 3$, $a_2 = -2$, $a_3 = 1$ and $a_4 = 0$. Compute each of the summations:

$$(a) \sum_{i=0}^4 a_i \quad (b) \sum_{j=0}^2 a_{2j} \quad (c) \sum_{k=1}^1 a_k$$

SOLUTION:

$$(a) \sum_{i=0}^4 a_i = a_0 + a_1 + a_2 + a_3 + a_4 \\ = 2 + 3 + (-2) + 1 + 0 = 4$$

$$(b) \sum_{j=0}^2 a_{2j} = a_0 + a_2 + a_4 \\ = 2 + (-2) + 0 = 0$$

$$(c) \sum_{k=1}^1 a_k = a_1 \\ = 3$$

EXERCISE:

Compute the summations

$$1. \sum_{i=1}^3 (2i - 1) = [2(1) - 1] + [2(2) - 1] + [2(3) - 1] \\ = 1 + 3 + 5 \\ = 9$$

$$\begin{aligned} 2. \quad \sum_{k=-1}^1 (k^3 + 2) &= [(-1)^3 + 2] + [(0)^3 + 2] + [(1)^3 + 2] \\ &= [-1 + 2] + [0 + 2] + [1 + 2] \\ &= 1 + 2 + 3 \\ &= 6 \end{aligned}$$

SUMMATION NOTATION TO EXPANDED FORM:

Write the summation $\sum_{i=0}^n \frac{(-1)^i}{i+1}$ to expanded form.

SOLUTION:

$$\begin{aligned} \sum_{i=0}^n \frac{(-1)^i}{i+1} &= \frac{(-1)^0}{0+1} + \frac{(-1)^1}{1+1} + \frac{(-1)^2}{2+1} + \frac{(-1)^3}{3+1} + \dots + \frac{(-1)^n}{n+1} \\ &= \frac{1}{1} + \frac{(-1)}{2} + \frac{1}{3} + \frac{(-1)}{4} + \dots + \frac{(-1)^n}{n+1} \\ &= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{(-1)^n}{n+1} \end{aligned}$$

EXPANDED FORM TO SUMMATION NOTATION:

Write the following using summation notation:

$$1. \quad \frac{1}{n} + \frac{2}{n+1} + \frac{3}{n+2} + \dots + \frac{n+1}{2n}$$

SOLUTION:

We find the kth term of the series.

The numerators forms an arithmetic sequence 1, 2, 3, ..., n+1, in which

$$a = \text{first term} = 1$$

& $d = \text{common difference} = 1$

$$a_k = a + (k - 1)d$$

$$= 1 + (k - 1)(1) = 1 + k - 1 = k$$

Similarly, the denominators forms an arithmetic sequence

n, n+1, n+2, ..., 2n, in which

$$a = \text{first term} = n$$

$$d = \text{common difference} = 1$$

$$\begin{aligned} \therefore a_k &= a + (k - 1)d \\ &= n + (k - 1)(1) \\ &= k + n - 1 \end{aligned}$$

Series

Hence the k th term of the series is

$$\frac{k}{(n-1)+k}$$

And the expression for the series is given by



$$\begin{aligned}\therefore \frac{1}{n} + \frac{2}{n+1} + \frac{3}{n+2} + \cdots + \frac{n+1}{2n} &= \sum_{k=1}^{n+1} \frac{k}{(n-1)+k} \\ &= \sum_{k=0}^n \frac{k+1}{n+k}\end{aligned}$$

TRANSFORMING A SUM BY A CHANGE OF VARIABLE:

Consider $\sum_{k=1}^3 k^2 = 1^2 + 2^2 + 3^2$

and $\sum_{i=1}^3 i^2 = 1^2 + 2^2 + 3^2$

Hence $\sum_{k=1}^3 k^2 = \sum_{i=1}^3 i^2$

The index of a summation can be replaced by any other symbol. The index of a summation is therefore called a dummy variable.

EXERCISE:

Consider

$$\sum_{k=1}^{n+1} \frac{k}{(n-1)+k}$$

Substituting $k = j + 1$ so that $j = k - 1$

When $k = 1$, $j = k - 1 = 1 - 1 = 0$

When $k = n + 1$, $j = k - 1 = (n + 1) - 1 = n$

Hence

$$\begin{aligned}\sum_{k=1}^{n+1} \frac{k}{(n-1)+k} &= \sum_{j=0}^n \frac{j+1}{(n-1)+(j+1)} \\ &= \sum_{j=0}^n \frac{j+1}{n+j} = \sum_{k=0}^n \frac{k+1}{n+k} \quad (\text{changing variable})\end{aligned}$$

Transform by making the change of variable $j = i - 1$, in the summation

$$\sum_{i=1}^{n-1} \frac{i}{(n-i)^2} \quad **$$

PROPERTIES OF SUMMATIONS:

$$1. \sum_{k=m}^n (a_k + b_k) = \sum_{k=m}^n a_k + \sum_{k=m}^n b_k; \quad a_k, b_k \in \mathbb{R}$$

$$2. \sum_{k=m}^n c a_k = c \sum_{k=m}^n a_k \quad c \in \mathbb{R}$$

$$3. \sum_{k=a-i}^{b-i} (k+i) = \sum_{k=a}^b k \quad i \in \mathbb{N}$$

$$4. \sum_{k=a+i}^{b+i} (k-i) = \sum_{k=a}^b k \quad i \in \mathbb{N}$$

$$5. \sum_{k=1}^n c = c + c + \dots + c = nc$$

EXERCISE:

Express the following summation more simply:

SOLUTION:

$$\begin{aligned} & 3 \sum_{k=1}^n (2k-3) + \sum_{k=1}^n (4-5k) \\ & 3 \sum_{k=1}^n (2k-3) + \sum_{k=1}^n (4-5k) \\ & = 3 \sum_{k=1}^n 3(2k-3) + \sum_{k=1}^n (4-5k) \\ & = \sum_{k=1}^n [3(2k-3) + (4-5k)] \\ & = \sum_{k=1}^n (k-5) \\ & = \sum_{k=1}^n k - \sum_{k=1}^n 5 \\ & = \sum_{k=1}^n k - 5n \end{aligned}$$

ARITHMETIC SERIES:

The sum of the terms of an arithmetic sequence forms an arithmetic series (A.S). For example

$$1 + 3 + 5 + 7 + \dots$$

is an arithmetic series of positive odd integers.

In general, if a is the first term and d the common difference of an arithmetic series, then the series is given as: $a + (a+d) + (a+2d) + \dots$

SUM OF n TERMS OF AN ARITHMETIC SERIES:

Let a be the first term and d be the common difference of an arithmetic series. Then its n th term is:

$$a_n = a + (n-1)d; \quad n \geq 1$$

If S_n denotes the sum of first n terms of the A.S, then

$$S_n = a + (a+d) + (a+2d) + \dots + [a + (n-1)d]$$

$$= a + (a+d) + (a + 2d) + \dots + a_n$$

$$= a + (a+d) + (a + 2d) + \dots + (a_n - d) + a_n \dots\dots\dots(1)$$

where $a_n = a + (n - 1) d$

Rewriting the terms in the series in reverse order,

$$S_n = a_n + (a_n - d) + (a_n - 2d) + \dots + (a + d) + a \dots\dots\dots(2)$$



Adding (1) and (2) term by term, gives

$$2 S_n = (a + a_n) + (a + a_n) + (a + a_n) + \dots + (a + a_n) \quad (n \text{ terms})$$

$$2 S_n = n (a + a_n)$$

$$\Rightarrow S_n = n(a + a_n)/2$$

$$S_n = n(a + l)/2 \dots\dots\dots(3)$$

Where

Therefore

$$l = a_n = a + (n - 1)d$$

$$S_n = n/2 [a + a + (n - 1) d]$$

$$S_n = n/2 [2 a + (n - 1) d] \dots\dots\dots(4)$$

EXERCISE:

Find the sum of first n natural numbers.

SOLUTION:

Let $S_n = 1 + 2 + 3 + \dots + n$

Clearly the right hand side forms an arithmetic series with

$$a = 1, \quad d = 2 - 1 = 1 \quad \text{and} \quad n = n$$

$$\therefore S_n = \frac{n}{2} [2a + (n-1)d]$$

$$= \frac{n}{2} [2(1) + (n-1)(1)]$$

$$= \frac{n}{2} [2 + n - 1]$$

$$= \frac{n(n+1)}{2}$$

EXERCISE:

Find the sum of all two digit positive integers which are neither divisible by 5 nor by 2.

SOLUTION:

The series to be summed is:

$$11 + 13 + 17 + 19 + 21 + 23 + 27 + 29 + \dots + 91 + 93 + 97 + 99$$

which is not an arithmetic series.

If we make group of four terms we get

$$(11 + 13 + 17 + 19) + (21 + 23 + 27 + 29) + (31 + 33 + 37 + 39) + \dots + (91 + 93 + 97 + 99) = 60 + 100 + 140 + \dots + 380$$

which now forms an arithmetic series in which

$$a = 60; \quad d = 100 - 60 = 40 \quad \text{and} \quad l = a_n = 380$$

To find n , we use the formula

$$a_n = a + (n - 1) d$$

$$\Rightarrow 380 = 60 + (n - 1) (40)$$

$$\Rightarrow 380 - 60 = (n - 1) (40)$$

$$\Rightarrow 320 = (n - 1) (40)$$

$$\frac{320}{40} = n - 1$$

$$\begin{aligned}8 &= n - 1 \\ \Rightarrow n &= 9\end{aligned}$$

Now

$$\begin{aligned}S_n &= \frac{n}{2}(a + l) \\ \therefore S_9 &= \frac{9}{2}(60 + 380) = 1980\end{aligned}$$

GEOMETRIC SERIES:

The sum of the terms of a geometric sequence forms a geometric series (G.S.). For example

$$1 + 2 + 4 + 8 + 16 + \dots$$

is geometric series.

In general, if **a** is the first term and **r** the common ratio of a geometric series, then the series is given as: $a + ar + ar^2 + ar^3 + \dots$

SUM OF n TERMS OF A GEOMETRIC SERIES:

Let **a** be the first term and **r** be the common ratio of a geometric series. Then its nth term is:

$$a_n = ar^{n-1}; \quad n \geq 1$$

If S_n denotes the sum of first n terms of the G.S. then

$$S_n = a + ar + ar^2 + ar^3 + \dots + ar^{n-2} + ar^{n-1} \dots \dots \dots (1)$$

Multiplying both sides by r we get.

$$r S_n = ar + ar^2 + ar^3 + \dots + ar^{n-1} + ar^n \dots \dots \dots (2)$$

Subtracting (2) from (1) we get

$$S_n - rS_n = a - ar^n$$

$$\Rightarrow (1 - r) S_n = a (1 - r^n)$$

$$\Rightarrow S_n = \frac{a(1 - r^n)}{1 - r} \quad (r \neq 1)$$

